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SUBSECTION 4.1 GENERAL REQUIREMENTS

4.1.1. SCOPE OF WORK

The Bidder shall erect and commission all the equipment furnished by him as required by the Contract in accordance with the requirements specified herein in **Part 4 Section 4 "Erection and Commissioning"**.

The Works shall conform to the applicable requirements of governing codes and all Indonesian codes or regulations having jurisdiction.

The Bidder shall arrange the schedule of his work in every aspect to meet the Contract Schedule.

The Bidder shall provide erection and commissioning procedures for all equipment furnished by him. These shall be submitted after Contract Award.

The Bidder shall furnish all erection tools, machinery, testing devices, expendable supplies, labour and labour supervision required and as necessary to properly erect and commission the equipment as specified in **Part 4, Subsection 1.2**

All workmanship shall be the best of its kind. All labour shall be performed by men skilled in their particular line of work and to the full satisfaction of the Owner. All of the Works shall be erected complete, ready for commercial operation and to the full satisfaction of the Owner.

4.1.2. GENERAL

4.1.2.1 WORK AND SUPERVISION

The erection and commissioning of all equipment shall be done under the guidance of the manufacturer's representative (as applicable) or in accordance with manufacturer's recommendations. The equipment shall be erected, set, leveled, aligned and completed in conformance with his direction and to his complete satisfaction before equipment is put into service. The manufacturer's representative shall also guide and direct the placing of the unit in successful operation by Owner's personnel.

The Bidder shall provide all Works and materials required for erection, assembly, mounting, wiring, checking, testing and commissioning for all equipment, components, controls, instruments, accessories and appurtenances furnished by the Bidder; and all other equipment supplied by other Bidders and erected by this Bidder as specified and as generally described in **Part 4, Subsection 1.2, Summary of Work**.

Certain accessories and appurtenances may be located remotely from the main equipment as shown on the drawings. It shall be part of Bidder's Works to set and erect such accessories and appurtenances and install interconnecting piping and / or wiring furnished by the manufacturer as an integral part of the whole unit.

The Bidder shall perform all work, which is normally considered part of the Work whether or not such Work is fully listed in detail in the Bid Document or Contract. The words "erect", "install", "installation", "checking", "testing", "correction", and "adjustment" will be frequently used concerning the Work of Erection and Commissioning.

4.1.2.2 Work, Schedules and Coordination

It shall be the responsibility of Bidder to engage the proper trades for carrying on all phases of the Works harmoniously. Bidder may sublet any portion of the Works, as may be required. However, the subBidder shall be subject to the written approval of the Owner, prior to performance of any work by subBidder.

It shall be the responsibility of the Bidder to co-ordinate his installation activities with the material delivery schedule of his subBidder and the construction program so that a smooth and harmonious schedule may be maintained. It shall also be the responsibility of the Bidder to co-ordinate his installation activities with his subBidders.

The Bidder shall be responsible for any work done by his SubBidders.

4.1.2.3 Interfacing

The Bidder shall cooperate with sub Bidders on the Site under his direction in the erection and commissioning of equipment associated or interfaced with other Bidder's work.

The Bidder shall provide all necessary interfacing information and cooperate fully with subBidders to check for interferences to ensure that interfacing information furnished will be as required to erect and commission all the equipment to the full satisfaction of the Owner.

For interfacing with existing plant refer to **Part 4, Section 1.2**

4.1.2.4 Acceptable and Unacceptable Work

All poor, defective or condemned work including work condemned because of interference with other Work shall be remedied at the expense of Bidder and he shall not be reimbursed for any expense in connection with such repairs. The Bidder shall be responsible for the care, safety and maintenance of all the materials until acceptance of the installed materials by the Owner. The Owner will not pay for the removal or replacement of damaged or faulty material after it has been installed or if damaged prior to installation. The Owner shall make the final decision as to whether Work is acceptable or not.

4.1.3. EQUIPMENT

4.1.3.1 Foundation

The Bidder shall not place equipment bases on foundation pads and floors before the top surface of the foundation pad or floor area, where base is to be mounted, has been properly chipped off, roughened and cleaned for grout.

The Bidder shall chip away at least 10 mm of foundation surface to provide a good bonding surface for grouting. Roughing of foundation surface shall be acceptable to equipment manufacturer's representative, prior to grouting of equipment.

4.1.3.2 Rotating Equipment

Bidder shall be responsible for the correct setting, mounting and alignment of all equipment erected by him and shall maintain and submit to Owner's representative written records of alignment readings. For 200 kW and larger rotating equipment the Owner's representative will inspect the completion of each of the shimming, grouting, alignment and dowelling work. Bidder shall not proceed with the subsequent work and operation of the equipment until each inspection is completed.

Alignment of rotating equipment and its motor drive shall include furnishing of anchoring devices of equipment and drives to base plates where required. This shall also include any special balancing of rotating elements of equipment due to faulty balancing in manufacturer's shops, checking and adjusting of bearings, setting of proper clearances, establishing the specified rotor to stator air gap for motors, realigning of all equipment which is shipped as a complete unit from the shop and rechecking of all alignment before startup. If equipment is realigned dowel holes shall be reamed again and the dowels reinstalled. Equipment shall not be operated without the dowels firmly set in place. Tolerance on initial shaft alignment and realignment of the above work shall be within the manufacturer's specified limits. Bidder shall drill equipment baseplates for drives where required.

The Bidder shall unload and erect the generator stator on the foundation as soon as possible after stator delivery to the project site. The erection method shall be as submitted to and accepted by the Owner.

4.1.3.3 Preservatives

The Bidder shall be responsible during the installation process for the complete removal of all preservatives applied to power plant equipment components which are incompatible with the system path and auxiliary systems to which they are applied.

The Bidder shall provide any and all equipment, mineral spirits, or solvents required to completely remove all preservatives from the equipment parts.

The Bidder shall install components immediately after removal of preservatives. If this cannot be done Bidder shall apply a light duty preservative to the components which is compatible with the equipment. Light duty preservatives to be used shall be in strict accordance with the manufacturer's recommendations.

4.1.3.4 Cleaning

In each case the methods of cleaning and/or flushing shall be as approved by the equipment manufacturer and as reviewed by the Owner.

Oil and gland steam piping other than stainless steel oil piping furnished by the Bidder shall be pickled in the shop. All piping shall be thoroughly cleaned by the Bidder before it is installed and all lines shall also be thoroughly cleaned out after erection so as to remove any possibility of dirt or any other foreign material being left in the system when the plant equipment is placed in operation. The method of cleaning piping shall be subject to review and acceptance by the turbine manufacturer and the Owner. Any temporary piping and facilities used in conjunction with cleaning out the piping shall be furnished by the Bidder.

4.1.3.5 Heating and Blanketing

The Bidder shall supply, erect and maintain necessary heating equipment to keep electrical equipment free from moisture, in accordance with manufacturer's recommendations, for as long as necessary.

The Bidder shall maintain a nitrogen blanket system or other means in all heat exchangers as necessary and provide all maintenance operation and testing to ensure compliance with manufacturer's recommendations for storage and handling.

4.1.4. SUPERVISION AND MANUFACTURER'S REPRESENTATIVES

4.1.4.1 General

The Bidder shall supply all technical guidance and supervision required for the Work.

4.1.4.2 Manufacturer's Representatives

The Bidder shall furnish guidance from manufacturer's technical representative for erection and commissioning of the equipment under this Contract Document. A manufacturer's representative shall be included for each major equipment.

If the services of any additional manufacture's representatives are required by the Bidder, the total cost of such services shall be at the expense of the Bidder. All equipment covered by this Specification shall be unloaded, stored, protected, erected, set, and aligned in complete accordance with the manufacturer's instruction books or directives. All rework or additional work

required due to noncompliance with manufacturer's instructions shall be at the expense of the Bidder and no claims shall be made by Bidder for lack of understanding of the complete Scope of Work.

The Bidder shall identify his Site personnel including Erection Supervisors, Start-up Supervisors and Performance Test Supervisors. Such personnel shall be subject to approval by the Owner.

4.1.4.3 Records and Reports

The Bidder shall maintain a Site inspection record of Quality Assurance/Quality Control as specified in **Part 4, Subsection 2.7**.

Bidder's representative at Site shall maintain a daily log- book consisting of:

- Owner's instructions
- Owner's complaints
- Schedule of important events of the day
- Record of disagreements

4.1.4.4 Owner's Inspectors

The Owner will have his inspection personnel at Site, to inspect the work done by Bidder.

4.1.5. ERECTION

4.1.5.1 Equipment and Accessories

The Work shall be in accordance with the Specification and all work shall be done in orderly and workmanlike manner. It shall present a neat appearance when completed and shall conform to the best modern engineering practice for the class of work, notwithstanding any omission in the Drawings and Specifications.

Each piece of equipment shall be located, as shown on the drawings and shall be erected complete in all respects unless exceptions are specifically noted herein.

The Work shall include the assembly and alignment of components where equipment is shipped unassembled or knocked down, or where the components are shipped separately.

The Works shall include the erection of all piping, wiring, instruments, controls, drivers and other components and accessories furnished with the equipment, unless otherwise indicated in the Specification or on the drawings.

The erection of equipment shall be in accordance with the recommendations of the equipment manufacturers.

During the course of erection, the Owner shall have full access at all times for inspecting work progress and witnessing checks for accuracy as may be required.

Equipment shall be installed, strictly in accordance with dimensions or drawing, with minimum tolerances, particularly with respect to level of concrete floors, clearances, welding thicknesses etc. Bidder shall provide full details of all erection tolerances as required by Owner for Owner's approval.

The Bidder shall provide all erection check lists for the works, copies of which are to be provided to the Owner after signing at each stage of inspection.

Test for Civil Works.

Tests for Civil Work, including but not limited, the following:

- Concrete Works
- Earth Works
- Piling.

The Test shall be executed by the Bidder according to the requirement in the related Technical Specifications for each Civil Works.

4.1.5.2 Construction Equipment and Safety

Bidder shall furnish and install temporary ladders and scaffolding he may require. All temporary stairways, ladders or platforms which may be required by Bidder shall be furnished and installed by Bidder at his expense and shall be removed by him after they have served their purpose at his expense.

Bidder shall furnish all hoisting, rigging cables, slings and other construction equipment (regardless of size and type) and supplies required for unloading, assembly and erection of equipment to be erected by Bidder. Attachments to building steel for hoisting and rigging shall be reviewed by the Owner.

Bidder shall erect all temporary barricades to isolate the block under construction and commission that may be required as approved by the Owner. All such barricades shall be arranged so as to ensure the safety of the workmen and passersby. Bidder shall remove such protection at the completion of the Works.

The Bidder shall keep the construction Site as clean as possible in order to facilitate construction work and to avoid damaging the equipment and injuring workers. Piping flushing oil, chemicals, etc. used for equipment cleaning shall be properly disposed of by Bidder.

4.1.6. ADJUSTMENT AND CORRECTION WORK

At the completion of erection of each system, Bidder shall perform all the necessary tests, including hydrostatic tests and performance tests.

The Works shall be subject to inspection and acceptance by the Owner.

Bidder shall make such adjustments and do such other work as may be required by the Owner to place each piece of equipment erected by Bidder in proper operating condition ready for service. This adjustment and correction work shall be done by Bidder without additional cost to the Owner.

Any correctional work required to be done by Bidder in order to install the equipment properly and which is due to the manufacturers' faulty shop work or design shall be reported by Bidder to the Owner and shall be done by Bidder only after review by the Owner of the correctional procedures. This in no way shall effect the warranty on the equipment.

4.1.7. START UP AND COMMISSIONING

4.1.7.1 Preparation and Specialties

Prior to start up, the Bidder shall provide cleaning and flushing as required by each equipment.

Temporary pumps, piping, hose, valves, jumpers etc. not furnished with the equipment but required for start-up and commissioning shall be furnished by Bidder and removed after cleaning operations are completed, prior to performance testing.

4.1.7.2 Schedules

Upon the completion of erection the Bidder shall carry out the start up and commissioning of each equipment in accordance with the start-up and commissioning procedures as reviewed and approved by the Owner.

For Block 2, start-up and commissioning activities will be performed for the complete combined cycle. The Taking Over Certificate for combined cycle will be issued after successful completion of the Reliability Run and Performance Test. The warrantee period for the combined cycle will be started after issuing of the Taking Over Certificate.

For Block 3 & 4, start-up and commissioning activities will be performed for the complete combined cycle. The Taking Over Certificate for combined cycle will be issued after successful completion of the Reliability Run and Performance Test. The warrantee period for the combined cycle will be started after issuing of the Taking Over Certificate.

The startup and commissioning activities shall in general follow the sequence of events shown in **Exhibit 1**.

4.1.7.3 Operation and Supervision

From initial start-up until takeover by the Owner, the Bidder shall be responsible for the running, operation and maintenance of the Works.

The start up shall be supervised by the Bidder using the Owner's operators. The Owner's operators shall be trained by the Bidder to his satisfaction to ensure safe and orderly plant start up and operation. Under the Bidder's supervision the Owner's operators shall exclusively operate all controls including but not limited to the following:

- All controls located on the Operator Station in the central control room.
- All other controls located in the central control room.
- All controls located on the mimic control panel or the Operator Station
- All local controls from local panels for extension 500 kV switchyard in the existing substation control room.
- All controls and switchgear located in the electrical Room, DC and UPS rooms.

4.1.8. CHECKING AND TESTING

4.1.8.1 Scope of Work

The Bidder shall perform all specified testing required for the equipment erected or furnished by him. All testing shall be in accordance with the specified Codes and Standards and instructions and recommendations of manufacturers as specified at **Part 4, Subsection 2.6 "Performance Testing, Guarantees and First Inspection"** and individual Technical Specifications.

In addition to the tests listed above, the Bidder shall demonstrate the function and capability of all equipment and systems supplied under the Contract, to perform in accordance with the Specification requirements and applicable standards.

All tests to be performed at Site shall be agreed with the Owner prior to commencement of the Work. The Bidder shall prepare the details of the test schedule and submit these to the Owner for approval. The Owner will mark in the appropriate spaces his intention to attend or waive the invited tests or inspections.

A minimum of twenty one days notice of the readiness of plant for test or inspection shall be provided to the Owner by the Bidder.

The Bidder shall furnish the Owner all test data taken for tests performed by the Bidder's representative. The test data forms used for tabulating the information required shall be furnished by the Bidder as soon as possible after tests have been carried out, for the approval of the Owner.

Further information is given on this subject in the succeeding subsections.

4.1.8.2 Performance Testing

A Performance Test shall be conducted in the presence of Manufacturer's representative as applicable to ensure that the testing is conducted in

accordance with the Manufacturer's recommendations, and is consistent with the procedures in the operating manuals.

The testing shall be performed under Bidder's direction and Owner's supervision as explained above using the Owner's operators. At least one year prior to the scheduled commercial operating date the Bidder shall advise the Owner of the training requirements for the Owner's start-up and testing crew. Additional training requirements are specified in **Part 4, Subsection 2.13**.

The complete Combined Cycle performance tests for Block 2, 3 & 4 shall be refer to Schedule 6, Plant Performance Guarantees (Block 2 Combined Cycle Operation & Block 3 and Block 4 Add-On Steam Turbine Performance and Complete Combined Cycle).

4.1.8.3 Test Coordination and Result

If, in the opinion of the Owner, test results show improper adjustment, operation or performance of any equipment, the Bidder shall furnish all labour, equipment and materials required to remedy the situation to the satisfaction of the Owner and at no increase in cost to the Owner.

Further information is given on this subject in the succeeding subsections.

4.1.9. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality Control shall conform to the requirements of **Part 4, Subsection 2.7**.

The quality control procedures shall include, but not be limited to, the following:

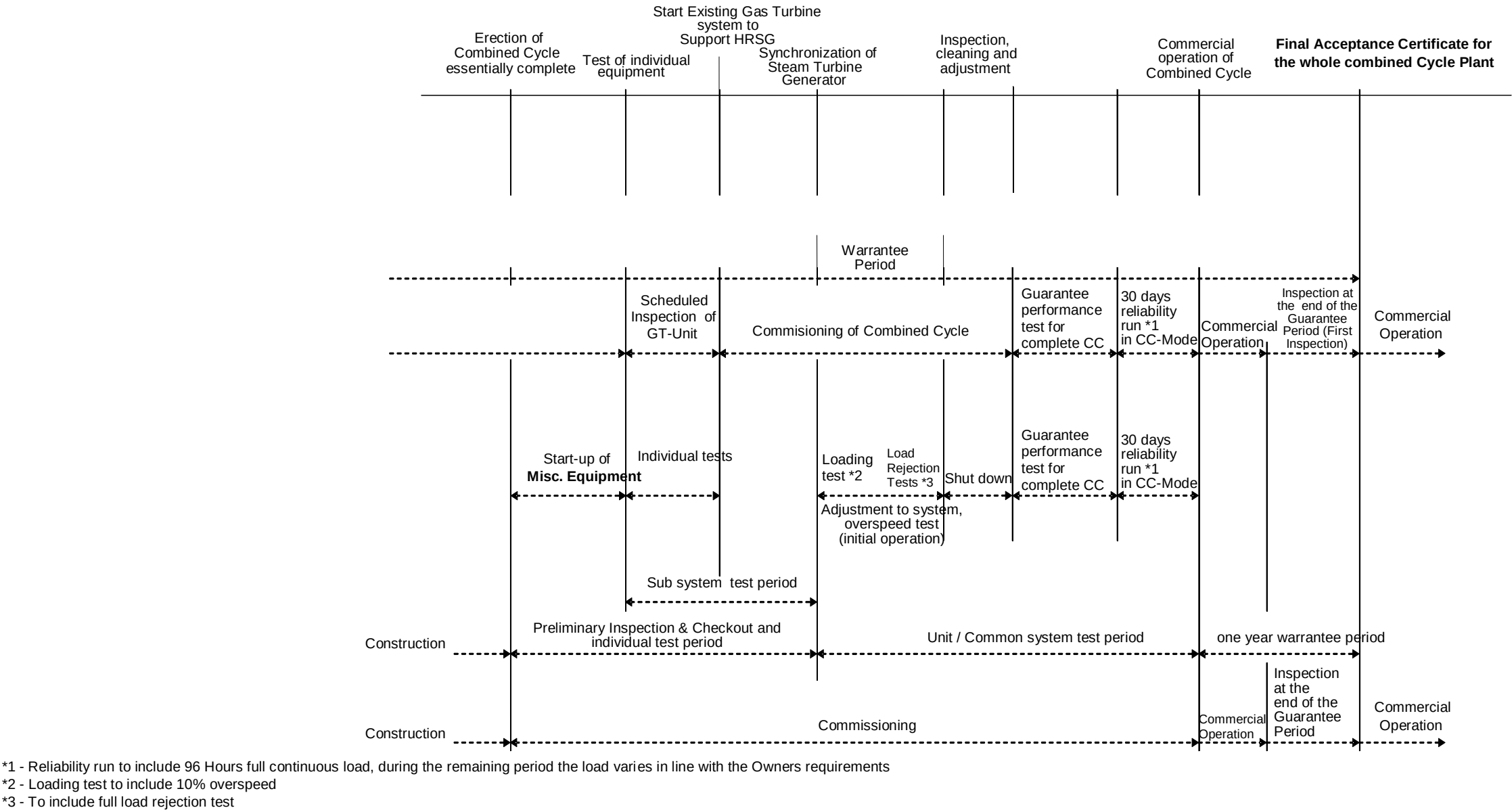
- Welding Procedure and Welding Procedure Qualifications.
- Radiography.
- Magnetic Particle Testing.
- Liquid Penetrant Testing.
- Ultrasonic Testing.
- Hydrostatic Testing and/or Pneumatic Testing.
- Leak Testing.
- PW Heat Treatment.
- Major Weld Repairs.
- Control of Welding Materials.
- Sub Bidder Qualification and Surveillance.
- Nonconformance and Deviation Control.
- NDE Personnel Qualifications.
- Wiring Insulation Testing.

- Control and Functional Testing.

The quality control documentation shall include, but not be limited to, the following:

- All documentation specified in the Specifications.
- All other documentation required by the applicable codes and standards.
- All other documentation required by the equipment manufacturer's field erection procedures.
- Material test reports.
- Material certification reports.
- Inspection reports.

Exhibit 1
Sequence of Events for Start Up and Commissioning for Combined Cycle Block 2, 3 & 4



(Last page of 4.1)

FOR REFERENCE